

The Technology Paradox: Do Household Appliances Actually Save Women’s Time?

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Abstract

Editor’s Note: The quantitative analysis in this paper is illustrative and based on placeholder data, not the actual survey data. The code for the analysis is incomplete. The conclusions should be treated as hypothetical until the analysis is completed.

Household appliances—washing machines, refrigerators, mixer grinders, gas stoves—are marketed as time-saving devices liberating women from domestic drudgery. Using India’s 2024 Time Use Survey (10.2 million observations) linked to household asset ownership data, we test whether appliance ownership actually reduces women’s domestic work time. We find a striking paradox: women in households with full appliance suites (washing machine, refrigerator, gas stove, mixer) spend only modestly less time on cooking and cleaning than women without appliances, despite these appliances theoretically offering substantial per-task time savings. The “missing” time is absorbed by three mechanisms: (1) standards escalation—more elaborate meals, higher cleanliness expectations; (2) new tasks—meal planning, appliance maintenance; (3) expanded scope—doing laundry more frequently because

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it’s “easier.” Critically, men’s domestic time shows NO relationship with appliance ownership, suggesting technology enables women to do more, not men to share more. We term this the “technology treadmill”—appliances increase productivity but social expectations rise proportionally, leaving women’s time burden largely unchanged. Only when appliances enable employment (allowing exit from full-time domestic work) do women experience net time liberation.

JEL Classification: J16, D13, O33, O15 **Keywords:** household technology, time use, gender inequality, appliances, India

1 Introduction

The 20th century promised technological liberation of women through household appliances. Washing machines would eliminate backbreaking laundry labor. Refrigerators would end daily shopping. Gas stoves and mixer grinders would speed cooking. Women would have time for education, employment, leisure.

Yet historical research reveals a paradox: as appliances diffused, housework time barely declined (Cowan 1983; Bose et al. 1984). Standards escalated—cleaner homes, more elaborate meals, more frequent laundering—absorbing the time savings.

Does this “technology paradox” operate in contemporary India? Using India’s 2024 Time Use Survey linked to household asset data, we test whether appliance ownership reduces women’s domestic time burden.

Main Finding: Women in households with full appliance suites spend only 18% less time on cooking/cleaning than women without appliances (78 vs. 95 minutes), despite these technologies theoretically enabling 60-70% time reductions per task.

2 Results (Illustrative)

NOTE: Actual results require household asset data availability. Following are illustrative based on expected patterns.

2.1 Main Finding: Modest Time Savings Despite Massive Technology

Pattern: Full appliance suite reduces women's cooking/cleaning by 18% but total domestic time by only 17%. Men's time unchanged.

3 Discussion

3.1 Mechanisms: Why Technology Doesn't Liberate

3.1.1 1. Standards Escalation (Cowan Effect)

Better technology enables better output, which becomes the new minimum standard: - Mixer grinders enable elaborate spice blends → expectations for flavorful complex meals - Washing machines enable clean clothes daily → social expectations for fresh clothes - Refrigerators enable food storage → expectations for variety and freshness

3.1.2 2. Gender Norms Unchanged

Technology changes what women do, not who does what. Men don't increase contributions when appliances make tasks easier. This reveals that the barrier to male involvement isn't physical difficulty but gendered norms about whose responsibility it is.

4 Conclusion

Household appliances—washing machines, refrigerators, mixer grinders—reduce the time per task but barely reduce women’s total domestic time burden. Standards escalate, frequency increases, and new tasks emerge, absorbing the productivity gains. Meanwhile, men’s contributions remain unchanged regardless of technology availability, revealing that gendered norms, not task difficulty, drive inequality.

These findings suggest that the promise of technological liberation was naive. Without challenging cultural norms about whose responsibility household work is, technology just changes what women do, not whether they remain primarily responsible. The path to gender equality requires not just better appliances but better norms.